

Assignment # 2

Deadline- 30 April 2026

1. Define visual image interpretation. Describe basic elements of visual image interpretation in remote sensing with illustrations.
2. Explain the importance of scale in satellite image interpretation and discuss how changes in scale influence the identification and classification of geographic features. Define image interpretation keys and describe the major elements used in visual image interpretation, such as tone/color, texture, shape, size, pattern, shadow, site, and association. Give examples.
3. Describe the various types of image preprocessing techniques. What would be the advantage of geometrically correcting an image to geographic coordinates prior to further analysis and interpretation?
4. Discuss various image enhancement techniques used in digital image processing. Explain the importance of spatial filtering in image enhancement and describe how different filters can be applied to modify the spatial frequency characteristics of an image, with suitable examples.
5. Explain the commonly used digital image processing functions with appropriate illustrations. Furthermore, differentiate between supervised classification and unsupervised classification in image analysis, highlighting their principles, workflow, advantages, and limitations.
6. Export the location of School data of Kathmandu Valley from openstreetmap to QGIS. (i) Identify all schools located within a 2 km radius of the Department of Chemistry, Tribhuvan University. ii. Create a map layer and a corresponding database table for schools located within the boundaries of Kathmandu Metropolitan City. iii. Determine the number of schools situated within a 1 km buffer of the Ring Road. Develop a separate map layer and database table for these schools.

(Hint: To Download OSM Data

Go to

<https://www.overpass-turbo.eu/>

Click on Wizard

Search (if don't know search name, go <https://wiki.openstreetmap.org/> to know tag name)

amenity=school in Kathmandu

Build and run query

Export in required format

Add school data in QGIS

Two theme display (point and poly).

Convert poly to point using geometry tool - Centroid (will get center point of each poly)

Merge initial point feature and new point from poly using merge tool

To solve above questions, please use municipal boundary map and road network map.)